

**LIST OF FENCING EQUIPMENT ALONGWITH TECHNICAL SPECIFICATION FOR FENCING
ACADEMY SAINIK SCHOOL NAGROTA**

S.No	Nomenclature	Qty	Technical Specification
1.	FIE Fencing Piste	04 Nos	A fencing piste is the strictly defined rectangular strip where the sport of fencing occurs, measuring 17-18 metres long by 1.5 metres wide. It is marked with a center line, two en-garde lines where fencers start each bout, and warning zones near the rear limits. In competitive settings, the surface is typically made of conductive materials like aluminium or metallic mesh, which are grounded to a scoring machine to ensure hits to the floor are not registered as points. Fencers must maintain their footwork within these boundaries, as stepping off the side halts play, while retreating past the back line results in a point for the opponent.
2.	Fencing Apparatus complete 07 with remote battery pillars	04 Nos	Apparatus for 3 weapons, F.I.E. approved, also suitable for high level competitions. Scorer: height of the digits 13cm, legible over 55m. Chronometer and Match Counter: height of digits 9cm, legible over 35m. Red, Green, and White lights signaling hits: brightly lit, legible at 180° as the lights protrude from the front panel. Red and yellow cards, priority and lights signaling faults: visible over 25m.The Manufacturer should have supplied and installed the quoted product to some State / National / Government recognized / Federation Recognized tournament / Training Camp.
3.	Millennium Reel /Spool	08 Nos	High quality springs. The springs are made from a special top-quality steel strip. Longer lasting cable. The cable quality has been improved for a still greater lasting resistance. Furthermore the large diameter of the cable coil not only enlarges The cable camber, but also reduces the number of its bending. Therefore the cable has a longer lasting (both for the external boot and 3 internal wires). Strong stainless steel case. The case is sturdy and easy to clean. In addition, thanks to its flat, square shape, it is easy to carry and store. The top of the reel has flat surface area of 33 cm x allowing for advertising Space. The Manufacturer should have supplied and installed the quoted product to some State / National / Government recognized / Federation Recognized tournament / Training Camp.
4.	Spool Connecting Wire	08 Nos	Floor cable 14m long, with 3- Pin plugs. The Manufacturer should have supplied and installed the quoted product to some State / National / Government recognized / Federation Recognized tournament / Training Camp.
5.	Piste Connecting Wire	04 Nos	Piste -Apparatus /Signal Unit Cable 5 m long. The Manufacturer should have supplied and installed the quoted product to some State / National / Government recognized / Federation Recognized tournament / Training Camp.
6.	Epee weapon electric complete	18 Nos	Epee electric weapon (complete with French grip) is made from high-strength carbon steel. It includes a wired blade with electric point, aluminum guard, insulated pad, socket, pommel, and straight French grip handle. Available in sizes 0–5 (youth to adult), with right/left-hand configuration. The electrical system ensures precise touch registration via low-resistance wiring. The French grip provides extended reach and flexible hand positioning. Durable construction supports repeated training use, with optional colored finishes and modular replaceable components for maintenance and customization.

7.	Foil weapon electric complete	18 Nos	Fencing Foil Electric Weapon (Complete), is a standard competition/training foil designed for electronic scoring. It features a high-tensile carbon or maraging steel blade (~90 cm), total weapon length up to 110 cm, and weight below 500 gm. The blade is flexible with rectangular cross-section for safety and precision. It includes an electric point with spring-loaded tip (500 g activation force), internal wiring, and 2-pin or bayonet socket. Components include aluminum guard, insulated pad, grip (French or pistol), and pommel assembly. The system ensures low-resistance conductivity, durability, and compatibility with standard scoring apparatus.
8.	Sabre weapon electric complete	18 Nos	Sabre Electric Weapon Complete S2000 is designed for training and competition. It features a high-carbon steel S2000 blade (sizes 0–5; adult ~88 cm), compliant with stiffness regulations. The complete assembly includes insulated guard, guard pad, socket, pommel, and rubber/PVC grip. The electrical system uses conductive pathways through blade and guard for valid touch detection. Surface coatings (standard, colored, or gold anti-rust finish) enhance durability. Lightweight construction ensures balance and maneuverability, while modular components allow easy replacement and maintenance. Compatible with standard scoring apparatus and body cords.
9.	Epee blade electric complete	18 Nos	Épée Electric Blade Complete refers to a stiff-grade (D stiffness) blade designed for enhanced point control and durability. It is manufactured from high-elasticity carbon or spring steel, heat-treated and tempered for flexibility without permanent deformation. Standard length is up to 90 cm (size 5), with triangular cross-section. The blade comes pre-wired with insulated internal wiring, spring-loaded electric tip (750 g pressure), and threaded barrel. Surface finishes include polished, anti-rust, blue, or gold coatings. Compatible with 2-pin sockets, it suits French and pistol grips, ensuring reliable conductivity and long service life.
10.	Foil blade electric complete	18 Nos	Foil Electric Blade Complete (D-type, is a stiff-grade fencing blade manufactured from high-tensile carbon or maraging steel, heat-treated for durability and controlled flexibility. Standard blade length is up to 90 cm with rectangular cross-section and tapered profile for precision. The blade is pre-wired with insulated internal wiring, featuring a spring-loaded electric tip (500 g activation force) for accurate scoring. It supports 2-pin or bayonet sockets and is compatible with French or pistol grips. Surface finishes include polished, anti-rust, blue, or gold coatings. Designed for training and non-FIE competition use, it ensures reliable conductivity and long service life.
11.	Sabre blade electric complete	18 Nos	Sabre Electric Blade Complete S2000 is a non-FIE compliant fencing blade manufactured from high-carbon or spring steel, heat-treated for controlled flexibility and impact resistance. It follows S2000 stiffness standards with a maximum blade length of 88 cm (size 5 adult) and total weapon length up to 105 cm. The blade features a rectangular cross-section with a reinforced spine for durability. It includes an integrated electrical wiring system connected to a spring-loaded guard circuit for touch registration. The assembly comes with insulated guard, socket, grip, and pommel. Available in anti-rust, gold, or colored coatings, it ensures reliable conductivity, balance, and long service life for training and club-level competition use.

12.	Mask Epee	18 Nos	350 N Épée Mask is a CE EN 13567 Level 2 fencing mask designed for high-protection épée training and competition. It features a high- strength stainless steel wire mesh rated for ~1000N penetration resistance and a bib rated to 1600N impact force. The shell is formed from reinforced carbon steel mesh with anti-corrosion coating and painted finish (white or black). Interior includes multi-layer foam padding, moisture- wicking removable liner, and adjustable rear strap system. The bib is stitched with durable synthetic fabric for shock absorption. It includes rear safety strap, elastic harness, and reinforced chin/neck protection. Weight approx. 900–1200 g, designed for stability, ventilation, and FIE-style fencing compatibility.
13.	Foil Mask	18 Nos	A Foil Mask 350N with Electric Bib is an EN 13567 Level 2 fencing mask designed for high protection and electronic scoring. It features a stainless-steel mesh shell rated ~1000N penetration resistance and a bib rated 1600N impact force. The integrated electric lamé bib is made from conductive stainless-steel or copper- coated fibers, ensuring valid touch registration in foil fencing. The mask includes multi-layer shock- absorbing foam, removable washable liner, ventilation zones, and adjustable rear strap for secure fit. Reinforced throat and chin protection improve safety. Weight is typically 900–1200 g, ensuring balance, durability, and competition-grade performance.
14.	Sabre Mask	18 Nos	Sabre Electric Mask 350N is an EN 13567 Level 2 fencing mask designed for sabre fencing with integrated electrical system compatibility. It features a stainless-steel mesh shell tested for ~1000N penetration resistance and a 1600N- rated conductive bib made from stainless-steel or silver-coated conductive fabric to complete the scoring circuit. The interior includes multi-layer shock-absorbing foam, moisture-wicking removable liner, and reinforced forehead padding. An adjustable rear strap system ensures a secure fit during fast sabre actions. Ventilation zones improve airflow, while reinforced chin and throat protection enhance safety. Weight is typically 900–1200 g, suitable for training and competition use.
15.	Epee Body Wire	18 Nos	Épée Body Wire is an electrical signal transmission cable connecting the épée weapon to the reel system and scoring apparatus. It typically uses a 3-prong weapon plug (A, B, C lines) made of nickel-plated brass or stainless steel for stable conductivity. The cable is multi- strand copper core with PVC or TPU insulation for flexibility and abrasion resistance. Standard length is ~1.5–2.5 meters with reinforced strain relief at both ends to prevent wire breakage. It connects via bayonet or clip-lock reel plug. Designed for low resistance signal flow, it ensures accurate touch detection and stable circuit continuity in training and competition environments.

16.	Foil / Sabre Body Wire	36 Nos	A Foil & Sabre Body wire is an electrical connection cable linking the fencing weapon and lamé to the scoring apparatus via reel. It uses a 2-prong weapon plug with nickel-plated brass contacts for reliable conductivity and corrosion resistance. The internal core consists of multi-strand copper wires with flexible PVC or TPU insulation to withstand repeated bending during footwork. For foil, it connects blade circuit and conductive bib; for sabre, it carries continuous circuit through blade and target lamé. Standard length is 1.5–2.5 m with reinforced strain relief and secure reel-end connector, ensuring low resistance, durability, and stable signal transmission in training & competition use.
17.	Fencing Suit	54 Nos	A Fencing Suit typically consists of a jacket, breeches (pants), and optional underarm plastron, manufactured according to CE EN 13567 standards in 350N (Level 1) or 800N (Level 2/FIE-grade) protection levels. The suit is made from high-strength polyester, cotton-blend, or elastic technical fabric with reinforced multilayer construction for puncture resistance and mobility. The jacket includes front zip or strap closure, reinforced chest padding, and elastic cuffs for secure fit. Breeches are high-waist with adjustable suspenders, reinforced knee areas, and stretch panels for movement. Many models feature breathable lining for sweat absorption and comfort during long training sessions. suits are often modular, lightweight, and customizable in size and branding for clubs. Higher-end versions (800N) offer greater impact resistance and durability, suitable for competitions, while 350N suits are used for beginners and training. Weight ranges approx. 0.9–1.5 kg depending on level and material layering.
18.	Fencing Chest Protector	18 Nos	A fencing chest protector is typically constructed from high-impact resistant polycarbonate or polyethylene, offering lightweight yet durable protection. It features an ergonomic, contoured design to fit the torso comfortably without restricting movement. The surface is smooth to prevent blade snagging and complies with Fédération Internationale d'Escrime (FIE) safety standards. Thickness ranges from 2–3 mm, ensuring optimal shock absorption. Adjustable elastic straps or harness systems provide a secure fit. The protector is sweat-resistant, easy to clean, and available in men's and women's variants, with additional foam padding or gel inserts for enhanced comfort and impact distribution.
19.	Brest protector	18 Nos	Fencing breast protector is manufactured from high-density polycarbonate or polyethylene, ensuring strong impact resistance and durability. It is anatomically shaped for female athletes, providing full chest coverage with a contoured ergonomic fit. The surface is smooth and seamless to reduce blade friction and prevent snagging. Thickness typically ranges from 2–3 mm for effective shock absorption. It complies with Fédération Internationale d'Escrime (FIE) standards. The protector includes adjustable elastic straps or a harness system for a secure fit, along with optional soft foam or gel padding for added comfort, stability, and even distribution of impact forces during bouts

20.	Sabre Electric Jacket	18 Nos	Sabre electric jacket (lamé) is constructed from conductive metallic fabric, typically stainless steel or copper-nickel fibers woven with synthetic yarn for durability and flexibility. It covers the valid target area—torso, arms, and collar—and forms an electrical circuit to register touches, as defined by Fédération Internationale d'Escrime rules. The surface resistivity is usually below 5 ohms for reliable conductivity. It features a front or side zipper, conductive bib connection, and grounding clip points. Lightweight (approx. 500–800 g), it includes inner lining for comfort, anti-corrosion coating, and is available in multiple sizes with ergonomic fit.
21.	Foil Electric Jacket	18 Nos	Foil electric jacket (lamé) is constructed from conductive fabric using stainless steel or copper- nickel threads blended with synthetic fibers for flexibility and durability. It covers the valid target area—the torso and groin—and ensures accurate scoring through electrical conductivity as per Fédération Internationale d'Escrime (FIE) standards. Surface resistance is typically below 5 ohms. The jacket features a front or back zipper, secure body cord connection, and reinforced seams. Lightweight (approx. 400–700 g), it includes a breathable inner lining, anti-oxidation treatment, and is available in various sizes with an ergonomic, competition-fit design.
22.	Sabre electric Gloves	18 Nos	sabre electric glove is constructed from durable synthetic leather combined with conductive metallic fabric (copper-nickel or stainless steel fibers) on the cuff to ensure scoring continuity. The conductive cuff overlaps the sabre lamé, maintaining an unbroken electrical circuit as per Fédération Internationale d'Escrime (FIE) standards. The palm features anti-slip reinforcement for secure weapon grip, while the backhand includes breathable fabric for comfort. Surface resistance of the conductive area is typically below 5 ohms. The glove has an elastic wrist closure or Velcro strap, reinforced stitching, sweat resistance, and ergonomic design for flexibility and durability.
23.	Foil /Epee Gloves	36 Nos	Foil and Épée Gloves are constructed from high- quality synthetic or genuine leather, ensuring durability, flexibility, and secure grip. The palm is reinforced with anti-slip material and padding for weapon control, while the backhand uses breathable stretch fabric for comfort and ventilation. Gloves feature an extended cuff to overlap the jacket sleeve, preventing skin exposure. They comply with Fédération Internationale d'Escrime (FIE) standards for safety and performance. Stitching is reinforced for longevity, with elastic or Velcro wrist closure for a snug fit. Lightweight design enhances dexterity, and materials are sweat-resistant and easy to maintain.

24.	Sabre Mask Wire	18 Nos	Sabre mask wire is designed to provide reliable electrical continuity between the conductive mask and scoring system. It consists of multi-strand copper core wiring with flexible PVC or silicone insulation for durability and bend resistance. Standard length is approximately 30–40 cm with 4 mm or 2-pin connectors compatible with body cords. The resistance is typically below 1–2 ohms to ensure accurate signal transmission. It includes secure clips or crocodile connectors for firm attachment to the mask. Manufactured in compliance with Fédération Internationale d'Escrime (FIE) standards, it is lightweight, corrosion-resistant, and designed for repeated use.
25.	Master Jacket	01 Nos	Fencing Master Jacket is constructed from high- strength cotton or polyester blend fabric, often reinforced with Kevlar or ballistic nylon panels for enhanced durability. It provides 350N to 800N puncture resistance, meeting Fédération Internationale d'Escrime (FIE) safety standards. The design includes extra padding on the chest, arms, and shoulders to absorb repeated impacts during coaching sessions. It features a front zip with Velcro flap, adjustable cuffs, and extended length for full coverage. Breathable inner lining ensures comfort, while reinforced seams and ergonomic tailoring allow flexibility, mobility, and long-lasting performance under intensive training conditions.
26.	Master Arm Guard	01 Nos	Fencing Master Arm Guard is constructed from high-density foam padding combined with durable polyester or nylon outer fabric for impact resistance and longevity. It provides extended protection to the forearm and elbow during coaching sessions. The guard features ergonomic shaping for a secure fit without restricting movement. Multiple adjustable Velcro straps ensure firm fastening and easy wear. Reinforced stitching enhances durability under repeated use. The inner lining is soft, breathable, and sweat-absorbent for comfort. Lightweight construction reduces fatigue, while the outer layer is abrasion-resistant and easy to clean, making it suitable for intensive daily training environments.
27.	Master Mask	01 Nos	Fencing Master Mask is constructed from high- strength stainless steel mesh with reinforced bib and padding for maximum protection during coaching. It offers 1600N or higher resistance, complying with Fédération Internationale d'Escrime (FIE) safety standards. The mask features an extended, thicker bib to absorb repeated impacts and protect the neck area. Internal foam padding ensures comfort and shock absorption, while an adjustable rear strap system provides a secure fit. The frame is epoxy-coated for corrosion resistance, and the design allows adequate ventilation, visibility, and durability for intensive training use.

28.	Master Leg Guard	01 Nos	Fencing Master Leg Guard is constructed from high-density foam padding with a durable outer layer of polyester or nylon for impact resistance and abrasion protection. It is designed to protect the thigh and knee areas during intensive coaching sessions. The guard features an ergonomic, contoured shape for full coverage without restricting movement. Multiple adjustable Velcro straps ensure a secure and customizable fit. Reinforced stitching enhances durability under repeated impacts. The inner lining is soft, breathable, and moisture-absorbing for comfort. Lightweight construction reduces fatigue, while the exterior is easy to clean and suitable for regular training use.
29.	Master Hand Glove	01 Nos	Fencing Master Hand Glove is constructed from premium synthetic or genuine leather for durability and flexibility. It features reinforced padding on the fingers, knuckles, and backhand to absorb repeated blade impacts during coaching. The palm includes anti-slip material for a secure grip on the weapon. Extended cuff design provides additional wrist and lower forearm protection. The glove incorporates breathable fabric panels for ventilation and moisture control. Elastic or Velcro wrist closure ensures a snug, adjustable fit. Reinforced stitching enhances longevity, while the lightweight, ergonomic design allows full dexterity and comfort during prolonged training sessions.
30.	Coach Mask complete (With leather pad)	01 Nos	Coach Fencing Mask complete with leather pad is constructed from high-strength stainless steel mesh with an epoxy-coated finish for corrosion resistance. It provides enhanced protection with a reinforced bib and thick genuine or synthetic leather padding around the head for shock absorption during intensive coaching. The mask typically meets Fédération Internationale d'Escrime (FIE) safety standards, offering 1600N resistance. It features an ergonomic internal padding system, adjustable rear strap or spring mechanism for secure fit, and extended coverage for neck and sides. Ventilation design ensures visibility, airflow, and comfort during prolonged training sessions.
31.	Kit Bag -Small Trolley	18 Nos	Small Trolley Fencing Kit Bag is constructed from high-denier polyester or nylon (600D–1000D) for durability and water resistance. It features a compact, compartmentalized design with dedicated sections for weapons, mask, shoes, and accessories. The bag includes a lightweight telescopic aluminum handle and smooth-rolling polyurethane wheels for easy transport. Reinforced stitching and heavy-duty zippers ensure long service life. Internal straps secure equipment, while external pockets provide additional storage. Dimensions typically range from 80–100 cm in length. The bag is lightweight, abrasion-resistant, easy to clean, and designed for convenient travel and daily training use.

32.	Fencing Target Board	10 Nos	Fencing Target Board is constructed from high- density foam or layered synthetic materials designed to absorb repeated blade impacts without deformation. The surface is covered with durable vinyl or canvas, often printed with target markings for precision training. It is mounted on a rigid backing such as plywood or reinforced plastic for stability. Standard dimensions vary (typically 60–100 cm), with thickness ensuring adequate shock absorption. The board may include wall-mount brackets or a freestanding frame. It is lightweight, weather-resistant, and suitable for foil, épée, and sabre practice, offering durability, consistent rebound, and easy maintenance for regular use.
33.	Fencing Target Dummy	03 Nos	Fencing Target Dummy is constructed from high- density foam or molded polyurethane, designed to simulate the human torso for realistic training. The outer layer is covered with durable synthetic leather or vinyl for abrasion resistance. It features anatomically shaped contours to replicate valid target areas for foil, épée, and sabre practice. The dummy is mounted on a stable base or adjustable metal stand for height variation. Internal structure ensures impact absorption and long-term shape retention. Lightweight yet sturdy, it is easy to install and maintain, weather-resistant, and suitable for intensive daily training in indoor or outdoor environments.
34.	Fencing Tool Kit	01 Nos	Fencing Tool Kit includes essential maintenance tools for foil, épée, and sabre weapons. It typically contains precision screwdrivers, hex keys, tip setters, spring gauges, wire strippers, pliers, and spare screws and springs. Tools are manufactured from hardened stainless steel or chrome-vanadium for durability and corrosion resistance. The kit is housed in a compact, shock- resistant case with organized compartments. Components. Lightweight and portable, the kit ensures accurate adjustments, easy repairs, and long-term reliability during training and competition use.